

Replacing the external LLM reasoner

A feasibility and deployment study

Revision 4 ▪ 12 August 2026

| Prepared for the project supervisor

| Teacher model: self-hosted

Qwen3-8B-AWQ (vLLM)

| Evaluation data: 240 RFMiD images

Abstract

We were asked to obtain a lightweight model for the screening pipeline’s reasoning stage, requiring modest inference compute, exceeding 75% accuracy with precision at least equal to the incumbent, and to establish generalisability before any container image was produced. We report that the requirement is satisfied for the *triage* sub-task by a 3 KB logistic regression that attains accuracy and macro-precision of 1.000 at 0.14 ms per inference, and that this model has been deployed and verified in service. We further report that the *narrative* sub-task is not yet suitable for clinical enablement, and we set out the measurements supporting that judgement. We draw the supervisor’s attention to two findings that qualify the headline result: the triage task proved to be deterministic, so the perfect scores indicate an eliminated dependency rather than a modelling advance; and a quantisation projection made in our previous revision has been tested and found incorrect, which we correct here.

1. Objectives

The brief specified four conditions. We restate them and record where each is addressed.

Condition	Status	Evidence
A lightweight final model not requiring substantial inference compute	Met	3 KB of JSON weights; 0.14 ms per inference on CPU; the serving package depends only on the Python standard library (§3.1)
Alternative architectures to be explored	Met	Seven candidates evaluated on a common split; five attained 1.000 (§3.2)
Accuracy above 75% with precision at least equal to the incumbent	Met	Accuracy 1.000 and macro-precision 1.000, against a threshold of 0.75 and an incumbent precision of 0.492 (§3.2)
Generalisability established before a container image is produced	Met	Four independent tests conducted prior to the build (§3.3)

AN IMPORTANT QUALIFICATION

We wish to be explicit that the perfect scores reported above should not be read as evidence of a superior model. On the out-of-sample set, the triage decision of the 8B teacher was

identical to the disease classifier’s own `referral__priority` field in **160 of 160 cases**. The task is therefore deterministic, and the deterministic rules, a 3 KB linear model and an 8B language model are indistinguishable on it. The contribution of this work is the removal of a language-model dependency — with its associated latency, operating cost and clinical-data egress — from the decision path, rather than an improvement in predictive accuracy.

2. Method

The pipeline delegates two distinct tasks to an external language model, and we treated them separately throughout, as they are not comparable. **Triage** is a four-way classification over the disease classifier’s structured output. **Narrative generation** produces free text that no classifier can emit.

All figures reported below were obtained against a live, self-hosted Qwen3-8B-AWQ teacher on RTX A6000 hardware. Where a figure is derived arithmetically rather than measured, we state so explicitly.

Table 1: Evaluation design. Sample size remains the principal limitation on the narrative results, and we return to this in §6.

Evaluation	n	Design	Strength
Triage, out-of-sample	160	Unseen images, differing label distribution	Strong
Triage, cross-validation	80	5-fold × 5 seeds (25 fits)	Strong
Triage, emergency stress	80	Injected CRAO/AION cases	Synthetic, decisive
Narrative, variant sweep	24	Six variants, single code path	Limited
Narrative, faithfulness	24	Claim-level audit	Limited

3. Results: triage

3.1 Selected model

We selected `feat__logreg`, a multinomial logistic regression over 27 structured features derived from the classifier’s output. The artefact is 3 KB of JSON weights and the forward pass is a single matrix multiplication, so the serving path requires neither scikit-learn nor a deserialisation step at start-up. We verified that the exported weights reproduce the fitted estimator’s predictions on all 240 available cases before adopting them.

3.2 Comparison of architectures

Table 2: Candidate architectures on a common held-out split. Five of the seven attained 1.000; we selected the smallest and fastest.

Model	Acc.	Macro-P	Macro-F1	Size	p95	Gate
feat_logreg	1.000	1.000	1.000	3 KB	0.14 ms	Pass
feat_xgboost	1.000	1.000	1.000	0.42 MB	1.8 ms	Pass
rule_baseline	1.000	1.000	1.000	0	0.02 ms	Pass
cnn_triage	0.792	0.492	0.520	6.1 MB	161 ms	Fail

The image-based convolutional model underperformed for a diagnosable reason: it infers triage from pixels, whereas the teacher derives it from the classifier’s structured output. Supplying that structured vector to a small tabular model recovers the mapping directly.

3.3 Generalisability

We conducted four tests before producing any container image, in accordance with the brief.

Test	Result
Out-of-sample (train 80, test 160 unseen images, differing label mix)	Accuracy, macro-precision, F1 and κ all 1.000
Repeated stratified 5-fold cross-validation (25 fits)	1.000 $\pm$ 0.000
Learning curve	1.000 from 20 training cases onward
EMERGENCY stress test (80 injected CRAO/AION)	Recall and precision 1.000

3.4 Safety provision

No EMERGENCY case occurs anywhere in the training sample; consequently the learned model cannot emit that class. Escalation for sight-threatening codes is enforced by a deterministic override in the serving path, so a learned model cannot downgrade such a finding. As no observed case exercises this path, we cover it by unit test. The loader additionally refuses any artefact whose feature layout has diverged from the one recorded at export, and degrades to the deterministic rules on any failure rather than failing a screening.

4. Results: narrative generation

We distilled a 135M-parameter narrator and evaluated it at three precisions on the held-out split. Output is plain prose rather than JSON, since the triage fields are now supplied by the model in §3; this removes the parse-failure mode that rendered an earlier variant unusable, and generation succeeded in every case at every precision.

Table 3: Compact narrator, 24 held-out cases. Generation succeeds uniformly, so fidelity rather than reliability distinguishes the variants.

Precision	Size	Gen. rate	Omission	Quoted probs	Acuity drift
bf16	213.5 MB	1.000	0.000	0.000	0.250
int8	107.3 MB	37.5 s per case; evaluation impracticable			
nf4	54.2 MB	1.000	0.417	0.167	0.583

 GROUNDS FOR WITHHOLDING CLINICAL ENABLEMENT

The 4-bit variant substitutes a visible failure for an invisible one. It satisfies the 60 MB constraint and produces fluent text in every case, while omitting findings the teacher reported in 42% of cases. Fluent omission is considerably harder to detect in review than an absent output.

The bf16 variant’s clean scores do not extend beyond the evaluation distribution. The audit set is dominated by multi-finding cases resembling the training data. Presented with a single-finding case constructed by hand, the model reported “a 75% detection of diabetic retinopathy with a prevalence of 88%” for one finding at 88% confidence; both figures are fabrications. A single probe is not a measurement, but it demonstrates that the reported 0.000 is a property of the evaluation split rather than of the model.

4.1 Cost on the deployed hardware

The deployed Space runs on free cpu-basic hardware (2 vCPU). Measured there, restricted to two threads, the narrator requires 8.6 s to load, 426.9 MB resident, and 6–10 s per report, against a /predict endpoint that currently responds in approximately 1 s. We note additionally that transformers is not a declared dependency of the deployed image, so enabling the narrator would require an image rebuild, and that the smaller quantised variants cannot run on this tier at all, since the bitsandbytes kernels require CUDA. The narrator is therefore published but disabled by default.

5. Quantisation format study

Our previous revision recommended AWQ or GPTQ over bitsandbytes, and projected that GGUF Q4_K_M would reach approximately 80 MB. We have since tested the GGUF path and report that the projection was incorrect.

Table 4: The AWQ/GPTQ row remains arithmetic, as neither library could be installed in this environment. The GGUF rows are measured, using `convert_hf_to_gguf.py` and `llama-quantize` built from source.

Format	Body	Embedding	BPW	Total
bitsandbytes NF4	4-bit	fp16, not quantised	—	109.8 MB
bitsandbytes int8	int8	fp16, not quantised	—	107.3 MB
AWQ / GPTQ (W4A16)	4-bit	fp16, excluded by convention	—	~110 MB
GGUF Q4_K_M	mixed	q8_0	6.17	105.5 MB
GGUF Q5_K_M	mixed	q8_0	—	112.1 MB
GGUF Q8_0	q8_0	q8_0	—	144.8 MB

At 105.5 MB the GGUF build is only marginally smaller than the 109.8 MB bitsandbytes build, rather than the substantial reduction our arithmetic implied. We identified two architecture-specific causes and then tested the explanation directly.

1. The embedding table *is* quantised, confirming the principle we argued, but to q8_0 rather than 4-bit: 54.00 MiB to 28.69 MiB, a factor of 1.9 rather than the factor of 4 assumed.
2. The hidden size of SmolLM2-135M is 576, which is not divisible by 256 as the q4_K block structure requires. Consequently **180 tensors fell back** to larger types, and llama.cpp reports the result as 6.17 bits per weight against the 4.5 assumed.

5.1 Control experiment

To establish that the divisibility constraint, rather than GGUF itself, accounts for the discrepancy, we repeated the procedure on `pythia-160m`, whose hidden size of 768 is divisible by 256. The comparison isolates the effect.

Table 5: Control experiment. The only material difference between the two models for this purpose is the hidden dimension.

	SmolLM2-135M (576)	pythia-160m (768)
q4_K fallbacks	180	0
Bits per weight	6.17	5.33
Embedding quantised to	q8_0	q4_K
Embedding reduction	1.9×	3.6×

The control confirms the diagnosis: with a divisible hidden dimension the fallback does not occur, the embedding receives 4-bit treatment, and the encoding falls to 5.33 bits per weight. Applying that rate to our 134.5M-parameter narrator gives approximately **89.6 MB**, which would fall inside the 60–100 MB band. We therefore conclude that the projection was directionally sound but arithmetically optimistic, and that it fails for this model on account of its width rather than on account of the format.

5.2 Compatibility with vocabulary pruning

We confirmed that the vocabulary-pruned checkpoint cannot be converted to GGUF: the conversion script requires `vocab_size` to equal the tokeniser size, and the pruned model reports 920 against 49,152. Pruning and standard quantisation formats are therefore alternative strategies rather than complementary ones.

5.3 Outstanding measurement

Fidelity under GGUF remains unmeasured, as `llama-cli` did not return on this host even for a trivial prompt. Whether Q4_K_M avoids the 42% omission observed in the pruned 4-bit build is therefore still open. We note that at 105.5 MB it would in any case compete against the bf16 build at 213.5 MB rather than offer an edge-deployable option.

6. Deployment and verification

The platform is deployed to the Space `landwind22/Retinal_Screening` and is in service; `/health` reports `model_loaded: true` across all 45 disease classes. It is the single deployment target, the previous Space having been retired.

We record three defects in the deployment pipeline identified during this work, all since corrected. The workflow previously reported success upon completion of the git push, and on two occasions published a Space that subsequently failed to start — once through a README whose front matter was overwritten, and once through a Dockerfile removed by `rsync -delete`. A third deployment was lost when a push-triggered run and a manual dispatch executed concurrently and raced to push. The workflow now polls the runtime stage and requires `/health` to respond before reporting success, serialises runs, and retries a rejected push. A six-hourly keep-alive maintains the Space against its 48-hour inactivity timer.

Table 6: Mobile audit (Lighthouse, Moto G Power emulation, throttled 4G), representing the mid-range Android hardware this product targets.

	Before	After	
Performance	80	95	
Accessibility	96	100	
Largest Contentful Paint	5.0 s	2.6 s	target 2.5 s
Transferred payload	776 KB	293 KB	−62%
Colour-contrast failures	31	0	
Redundant alt attributes	2	0	

Two substantive defects were corrected. Text compression was not enabled, so 481 KB of a 776 KB payload was transmitted uncompressed at a cost of 2.4 s on a throttled connection; the remedy required `gzip_proxied any`, since `nginx` does not compress proxied responses by default and the payload in question is proxied. Colour contrast failed on 31 nodes, of which the largest group arose not from an unsuitable colour but from an opacity setting reducing a compliant colour to a ratio of 2.71:1. We recomputed each value against its actual background; we note that on dark sections the correction is lighter rather than darker, a distinction that produced one regression before we identified it.

Container image sizes are 1.01 GB compressed and 2.66 GB on disk for the CPU variant, and

5.90 GB compressed for the GPU variant, of which 5.3 GB is the CUDA runtime and the CUDA build of PyTorch.

7. Limitations

We ask the supervisor to weigh the following when interpreting this report.

1. The narrative evaluations rest on 24 held-out cases and 56 training examples. This is sufficient to separate a functioning model from a broken one and little else; confidence intervals on the reported proportions are correspondingly wide.
2. No clinician has reviewed the generated text. We regard this as the principal barrier to clinical enablement.
3. No EMERGENCY case occurs in the 240 available traces, so escalation behaviour is established only by injected stress testing.
4. The positive prediction path could not be exercised end-to-end against the deployment, as the fundus images in the repository are dangling symbolic links into a cleared cache. The rejection path was verified.
5. Lighthouse scores vary by approximately five points between runs on shared infrastructure; single values should not be read as trends.

8. Recommendations and further work

We recommend that the triage model be considered complete: it is validated, computationally negligible, and removes a language-model dependency from the decision path. We recommend that the grounded template remain the default narrative and that `NARRATOR_ENABLED` remain false, on the grounds that the available instrumentation cannot yet distinguish a sound narrator from an unsound one, which in a clinical screening context is the property that must be established first.

We propose the following sequence of further work:

1. Regenerate the teacher traces at scale with the AI-disclosure statement mandated, addressing the statistical-power and transparency limitations together. The teacher is self-hosted, so the marginal cost is time alone.
2. Construct a faithfulness evaluation of appropriate rigour — claim decomposition with entailment scoring — and obtain clinician review of a sampled set.
3. Should an edge-deployable narrator remain desirable, evaluate a base model whose hidden dimension is divisible by 256, as §5.1 indicates this is the condition under which the 60–100 MB band becomes attainable without vocabulary pruning.
4. Measure serving latency under vLLM before quoting any figure, as the present measurements are dominated by framework overhead rather than by the models.

Supporting material.

`docs/28-reasoner-cnn-vs-distilledqwen.md` — design and results

`docs/29-narrator-verification-and-gaps.md` — verification review

`outputs/reasoner_comparison_real/` `outputs/generalizability/`

Provenance. Measurements were obtained on RTX A6000 hardware against a live Qwen3-8B-AWQ teacher. Figures derived arithmetically rather than measured are identified as such at each occurrence.